COURSE OUTLINE

Faculty:	Faculty of Computing	**Page:**	1 of 7
Course code:	SECP2753	**Academic Session/Semester:**	20242025/2
Course name:	Data Mining	**Pre/co requisite (course name and code, if applicable):**	-
Credit hours:	3		

Course synopsis	This subject presents a comprehensive introduction to the understanding of knowledge discovery process in databases. Such methodological understanding is important to tackle projects of all sizes. A number of data mining techniques with its algorithms are explained. Students explore into the application of these techniques in both lab and industry. Students could apply the knowledge learnt to solve real world problems.			
Course coordinator (if applicable)	Dr Rozilawati binti Dollah @ Md Zain			
Course lecturer(s)	**Name**	**Office**	**Tel (07-55)**	**E-mail (@utm.my)**
	AP Dr Roliana Ibrahim	N28A-02-32	-	roliana@utm.my
	Dr Rozilawati binti Dollah @ Md Zain	N28A-02-29	-	rozilawati@utm.my

Mapping of the Course Learning Outcomes (CLO) to the Programme Learning Outcomes (PLO), Teaching & Learning (T&L) methods and Assessment methods:

No.	CLO	PLO	*Taxonomies and **generic skills	T&L methods	***Assessment methods
CLO1	Analyse the processes and techniques used in data mining.	PLO1 (KW)	C4	Lecture, active learning	Test (20%) Final Exam (30%)
CLO2	Produce data mining solutions for a given problem.	PLO3 (PS)	C3	Lab work	Assignment 1 (5%) Project 1 (5%) Project 2 (5%)
CLO3	Propose business opportunities and solutions in data mining.	PLO10 (ES)	BU2	Project-based learning	Assignment 2 (10%) Refection Project (5%) Project 1 (10%) Project 2 (10%)

Details on Innovative T&L practices:

No.	Type	Implementation
1.	Active learning	Conducted through in-class activities
2.	Project-based learning	Conducted through case study assignment. Tasks are given in sequential steps throughout the semester. Students in a group of 3 are given a case study that requires knowledge of data mining in identifying the potential business opportunities & the design of solutions. The report must comply with the case study specifications and be given in the form of a written report.

Prepared by: Name: PM Dr Roliana binti Ibrahim (Course Owner) Signature: Date: 14 March 2025	Certified by: Name: Dr Sharin Hazlin binti Huspi (Director) Signature: Date:

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Weekly Schedule:

Week 1 - 2: Online (17-28 March 2025)	**Topic 1: Introduction to Data Mining** - Advantages and reasons for data mining (DM) - Knowledge Discoveries in Databases (KDD) - CRISP-DM
Week 3 – 5: (31March - 18 April 2025) * Eidul-Fitri (31 March -1 April)	**Topic 2: Data Pre-processing (Announce of Group Project) & (Assignment1)** - Types of Data - Binary, Nominal and Categorical Variables - Numerical and Continuous Variables - Handling Missing Value and Redundancy - Detecting Outliers - Data Transformation - Dimensionality of Data Sets - Sampling and Unbalanced Datasets
Week 6 - 7: (21 April 2025 – 2 May 2025) *Labour Day (1 May)	**Topic 3: Classification (Lab 1)** Classification, Methodology and Performance Measures Classification and Prediction I, Analysis
Week 8 (5 May - 11 May 2025)	**MID-SEMESTER BREAK**
Week 9: (12-16 May 2025) *Wesak Day (12 May)	Classification and Prediction II, Analysis
Week 10 - 11: (19 – 30 May 2025)	**Topic 4: Clustering (Lab2)** Clustering 1 Clustering II, Visualisation (Lab) **(Test 1)**
Week 12 - 13: (2 - 13 June 2025) *Agong Birthday (2 June)	**Topic 5: Association Rules Mining (Assignment2)** Association Rules Mining I, Theories Association Rules Mining II, Analysis
Week 14: (16 - 20 June 2025)	**Project Demonstration**
Week 15: (23 - 27 June 2025)	**Lab Project Reflection Report**
Week 16: (30 June - 6 July 2025)	**Revision Week**

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Transferable skills (generic skills learned in course of study which can be useful and utilised in other settings):

Team working
Writing technical report

Student learning time (SLT) details:

Distribution of student Learning Time (SLT) Course content outline	Teaching and Learning Activities						TOTAL SLT
	Guided Learning (Face to Face)				Guided Learning Non-Face to Face	Independent Learning Non-Face to face	
CLO	**L**	**T**	**P**	**O**			
CLO 1	21h	10h			4h	20h	**55h (excluding 2h-T, 3h-F)**
CLO 2	6.5h				2.5h	9h	**18h**
CLO 3	14.5h	5h			2.5h	20h	**42h**
Total SLT	**42h**	**15h**			**9h**	**49h**	**115h**

Continuous Assessment		PLO	Percentage	Total SLT
1	Assignment 1 Assignment 2	PS ES	5 10	As in CLO2 & CLO3
2	Project 1 (Supervised Learning) Project 2 (Unsupervised Learning)	PS ES	15 15	As in CLO2 & CLO3
3	Project Reflection	ES	5	As in CLO3
4	Test	KW	20	**2h**
Final Assessment			Percentage	Total SLT
5	Final Examination	KW	30	**3h**
Grand Total SLT				**120h**

Special requirement to deliver the course (e.g: software, nursery, computer lab, simulation room):

Software: Excel, RapidMiner

Learning resources:

Text book (if applicable)

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Main references

J. Han and M. Kamber, Data Mining: Concepts and Techniques, 3 ed.: Academic Press, US, 2012.

I. H. Witten and E. Frank, Data Mining: Practical Machine Learning Tools and Techniques, 2 ed.: Elsvier, 2005.

Additional references
On-going

Online
http://elearning.utm.my

Academic honesty and plagiarism:

Assignments are individual tasks and NOT group activities (UNLESS EXPLICITLY INDICATED AS GROUP ACTIVITIES) Copying of work (texts, lab results etc.) from other students/groups or from other sources is not allowed. Brief quotations are allowed and then only if indicated as such. Existing texts should be reformulated with your own words used to explain what you have read. It is not acceptable to retype existing texts and just acknowledge the source as a reference. Be warned: students who submit copied work will obtain a mark of **zero** for the assignment and exams and disciplinary steps may be taken by the Faculty. It is also unacceptable to do somebody else's work, to lend your work to them or to make your work available to them to copy.

Other additional information (Course policy, any specific instruction etc.):

1. Attendance is compulsory and will be taken in every lecture session. Students with less than 80% of total attendance is not allowed to sit for final exam.
2. Students are required to behave and follow the University's dress regulation and etiquette all the time.
3. Exercises and tutorial will be given in class, and some may be taken for assessment. Students who do not do the exercise will lose the coursework marks for the exercise.
4. Assignments must be submitted on the due dates. Some points will be deducted for late submissions. Assignments submitted three days after the due date will not be accepted.
5. Make up exams will not be given, except to students who are sick and submit medical certificate which is confirmed by UTM panel doctors. Make up exam can only be given within one week of the initial date of exam.

No	Assessment	PLO1	PLO3	PLO10	TOTAL
		CLO1	CLO2	CLO3	
1	TEST 1	20			20
2	ASSIGNMENT 1		5		5
3	ASSIGNMENT 2			10	10
4	PROJECT 1		5	10	15
5	PROJECT 2		5	10	15
6	PROJECT REFLECTION			5	5
7	FINAL EXAM	30			30
TOTAL PLO		50	15	35	100

ONLINE LEARNING & TEACHING (L&T) PLAN

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REVIEW OF L&T ACTIVITIES TO INCLUDE ONLINE LEARNING						
Course learning outcome	**Guided Learning FTF hours (from CI)**	**Guided Learning FTF hours completed**	**Online Learning hours**			
			Activities	**Type of time spent**	**Estimated time**	**Total time**
CLO1: Analyse the processes and techniques used in data mining.	21		6 / 360 minutes			
			Live Interaction with students	The time spent in synchronous live interactions	180 minutes	180 minutes
			Students carry out 1 collaborative learning task in Breakout room based on the given exercise	Time spent for instructional activities	30 minutes	30 minutes
			Students write group findings on Padlet	Time spent for instructional activities	10 minutes	10 minutes
			Students spent time on averagely 10 screens for all the activities	The average time on ‘screen’ and the number of screens viewed.	5 mins x 10 screens	50 minutes
			Live Interaction with students to discuss students’ work	The time spent in synchronous live interaction	90 minutes	90 minutes

REVIEW OF ASSESSMENT PLAN					
Before (from CI)		**Revised***			
Continuous Assessment				Percentage	Total SLT
Summative Assessment				Percentage	

** All CLO should be maintained but changes can be made on the type of assessment. If changes have been made on the assessment type which affects the SLT, please adjust the SLT in the CI accordingly.*